upfront costs and providing a robust charging infrastructure needs to be the goal to attract more customers.

Battery - Answer to many problems

The charging infrastructure, an inseparable essence of building a robust EV ecosystem, is still in its nascent stage in the country. To put things in perspective, by the end of March 2021 India had 1,000 charging stations across different cities, which is roughly the same number of charging stations that China adds on a daily basis.

To counteract this dearth, improving the efficiency of battery technology has to be a necessary catalyst to keep the growth trend going. "In the interim, we feel that the improvement in battery technology will also be a game-changer for mass B2C adoption of electric two-wheelers, helping to circumvent the charging infrastructure development to an extent," advises Dewan.

Battery swapping, for instance, is gaining popularity for its ease of use and accessibility. "For the commuter segment, we expect battery swapping to be a concept that is going to evolve and become a more pragmatic solution, which will help simply because the usage of that segment is going to be significantly higher," informs Dewan.

Notably, batteries hold a much more important role than just being an alternative charging solution. "Battery is something that is at the heart of the electric vehicle and accounts for 35 to 40 percent of the total cost," says Naveen Munjal, MD, Hero Electric. According to recent projections by experts, battery costs are expected to lower considerably over the next decade as new chemistries are discovered, and localisation becomes the norm.

What is surprising is that India, despite all its hoo-hah for electric mobility, is yet to enter the realm of battery manufacturing, which is not only a lucrative opportunity but also essential for facilitating price parity in electric two-wheelers and increasing their adoption. Making it an even better deal is the production-linked incentive (PLI) scheme for manufacturing advanced chemistry cell (ACC) batteries approved by the Centre with an outlay of 180 million rupees.

Munjal agrees, "Localisation is a very important factor. There are some reports that say when battery manufacturing happens in India, it could be one of the lowest cost based manufacturers of lithium-ion batteries."

A different approach

Now, while a truckload of challenges continues to affect the industry, it has still undeniably presented various opportunities to make a mark in the segment. And even though manufacturing and R&D are crucial aspects of those opportunities, the electric revolution has also brought about the chance for startups and businesses to be creative with their business models, which was not possible with ICE vehicles.

The aggregator model, for one, has become a popular choice for new businesses, which helps in growing the industry in numerous ways. Dewan says that the aggregator model is a noteworthy creation of the industry that can help convert a large part of the ICE-using population into electric mobility users. "While B2B conversion is going to be a very important factor, its economics have already started to fall in place. What needs to fall in place now is the aggregator model."

The plethora of already thriving aggregator models, or any such model for that matter, is not alarming at all, as per Kedar Soman, Co-founder & CTO, eBikeGo. "By no means do I think there is any danger of competition. Right now, we are all scratching the surface of a big disruption or change that is happening."

As a matter of fact, as these parallel business models keep running, they are creating a certain synergy that is more focused on promoting electric mobility and proving their benefits and making it mainstream. The last-mile delivery strategy is an example of how innovative ideas are springing up at the face of this revolutionising technology. "Nobody has made delivery-specific vehicles in petrol vehicles before, and a lot of such kind of different innovations are going to happen," Soman says.

In fact, as e-commerce becomes more and more commonplace in Tier 2 and Tier 3 cities, the scope of growth for the aggregator business would be filled with abundance.

Future rides on indigenisation

India, being one of the biggest automobile markets, provides numerous opportunities for the EV market to flourish. India's huge automotive export potential, for instance, already gives indigenously manufactured electric two-wheelers and EV component makers an edge to enter various global markets. Along with that, there is also an opportunity to capture emerging markets.

"If you look at China and other Western countries, two-wheelers are not the mass market product over there. But if I look at emerging economies like Africa or Latin America, the product line is very similar to what we have here in India," says Abhishek Saxena, Public Policy Expert, NITI Aayog. This provides us with the first mover's advantage, enabling us to build our capacities for not only domestic markets but to also synergise it with the export market.

But riding this ambition on the back of localisation and domestic production is key, says Saxena.

"If we go for electrification, then in no way can we go for electrification with imported products." A crux of that initiative is to build products that are suited to serve the conditions of Indian roads and ecological conditions. Because electric is a new product, we need to adjust and build products which are conducive to our conditions of humidity, temperature, and so on.

"We have done a lot of manufacturing in India but not a lot of indigenous R&D patents are coming out of India," Saxena points out. "Until and unless you localise those contents, you will not be cost-competitive in that in the long run," adds Saxena.

Additionally, boosting efforts to complete the entire value chain is necessary if India wishes to come at par with China. EFY

This article is based on a panel discussion titled 'E2W India Market Forecast' hosted by Emobility+. The author, Siddha Dhar, is a business journalist at EFY